

Ex.No:3

Assessment of IT Project Risk Analysis using open-source tools

Aim:

To assess and analyze IT project risks using open-source risk analysis tools, identify potential risks, evaluate their likelihood and impact, and develop appropriate mitigation strategies to improve project success and reduce risk exposure.

Algorithm:

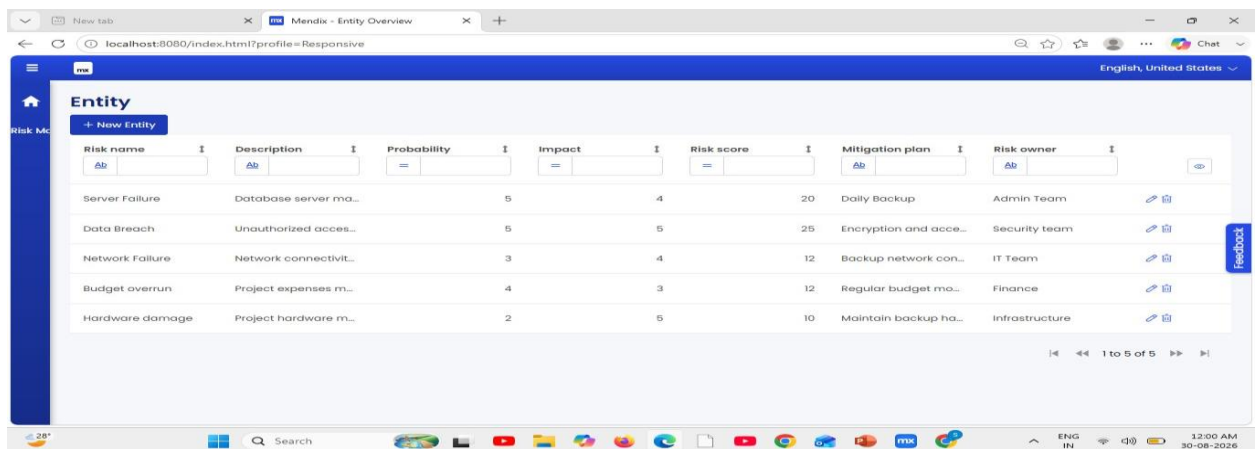
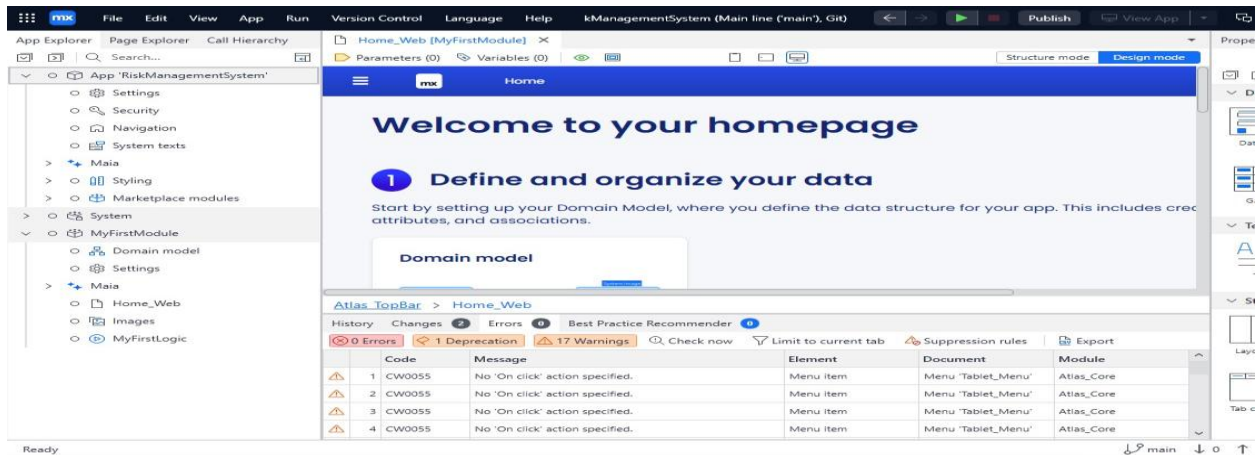
- Start the Mendix Studio Pro and create a new blank project named **RiskManagementSystem**.
- Create a **RiskManagement** module and open its Domain Model.
- Create a **Risk** entity to store project risk information.
- Add attributes such as **RiskName, Description, Probability, Impact, RiskScore, MitigationPlan, and RiskOwner**.
- Create **RiskCategory** and **RiskStatus** enumerations and add them to the Risk entity.
- Generate the **Risk Overview** and **Risk New/Edit** pages automatically.
- Configure the application navigation to provide access to the **Risk Overview** page.
- Create risk records and calculate the risk score using **Risk Score = Probability × Impact**.
- Test the **Create, Read, Update, and Delete (CRUD)** operations on the risk records.
- Verify the risk information and analyze the risks based on their **probability, impact, score, category, and status**, then stop.

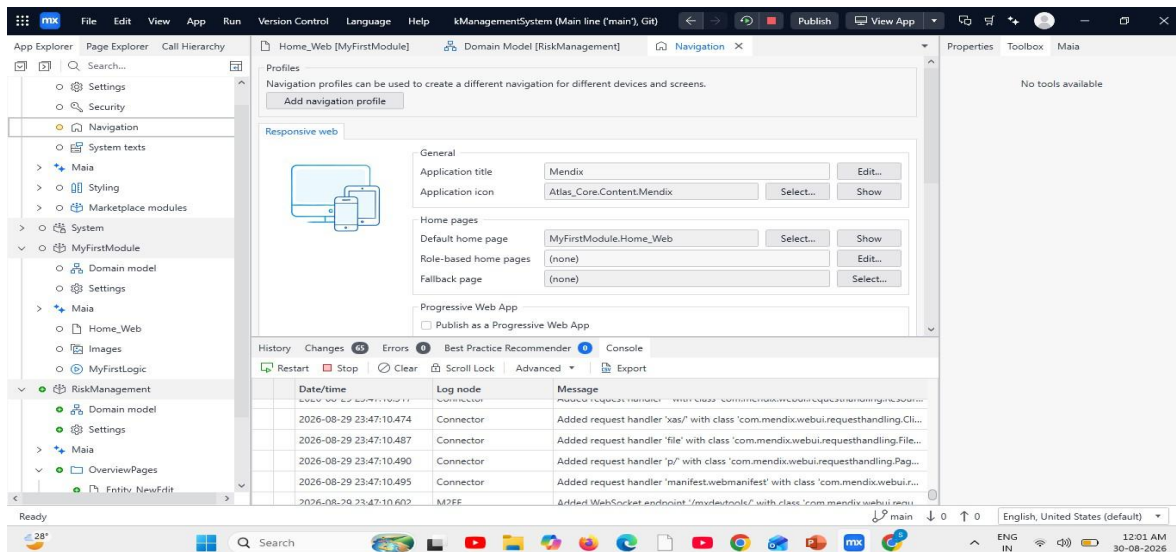
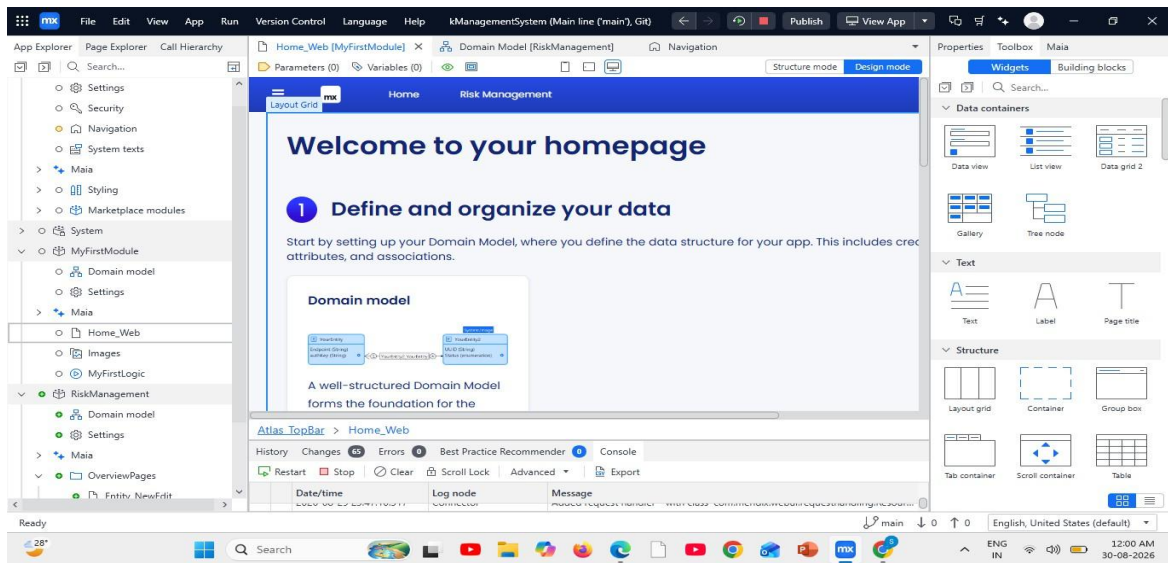
Procedure:

- Open Mendix Studio Pro, create a Blank App, and name it **RiskManagementSystem**.
- Add a module named **RiskManagement** and open the Domain Model.
- Create the **Risk** entity and add the required risk-related attributes.
- Create the **RiskCategory** enumeration with **Technical, Schedule, Cost, Resource, Security, and Quality** values.
- Create the **RiskStatus** enumeration with **Open, UnderReview, Mitigated, and Closed** values, and add both enumerations to the **Risk** entity.
- Generate the **Risk Overview** and **Risk New/Edit** pages from the **Risk** entity.

- Configure System → Navigation and add Risk Management to open the Risk Overview page.
- Run the application and create sample risk records with probability, impact, risk score, category, status, owner, and mitigation plan.
- Verify the records and test Create, Read, Update, and Delete operations.
- Confirm that all risk details and risk scores are displayed correctly and that the Risk Management System is functioning successfully.

Output:





Result:

The IT Project Risk Management System is successfully developed using Mendix, and the identified project risks can be recorded, assessed, monitored, and managed based on their probability, impact, risk score, and mitigation status.

